Journal of Energy Storage 35 (2021) 102273

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

A Brownian motion model to describe a random crystallization of undercooled water dispersed within emulsions

Hammou El-Otmany [a,*], Tarik El Rhafiki [a], Tarik Kousksou [b], Youssef Zeraouli [b]

[a] Engineering Sciences Laboratory, Polydisciplinary Faculty of Taza, Sidi Mohamed Ben-Abdellah University Fez, Morocco
[b] University of Pau, E2S UPPA, SIAME, 64000 Pau, France

ARTICLE INFO

Keywords:
Heat transfer
Crystallization
Stochastic modelling
Brownian motion
Genetic algorithm

ABSTRACT

This paper aims to propose a new mathematical model to describe heat transfers during stochastic crystallization of a metastable water dispersed inside an emulsion. This model is a combination between energy equation and nucleation laws. A finite volume method is chosen to solve the energy equation and a Brownian model with unknown parameters is introduced to explain the random behaviour of water crystallization within the emulsion. The inverse method is then used to estimate the unknown parameters for stochastic model by using the experimental temperature obtained from the literature. Finally, we present some numerical results to illustrate the inverse method and to validate our proposed stochastic model.

1. Introduction

1.1. Background and earlier work

Water emulsions are systems formed by two immiscible liquids in which one (dispersed phase i.e. water) is dispersed in another (emulsifying support). This dispersion takes place in the form of small water droplets [1]. Water emulsions are commonly manufactured in a two-stage process emulsification [2,3]. During the first stage, the dispersed phase is emulsified above its fusion temperature and crystallized during cooling in the second stage. The analysis of the crystallization of these droplets is of a major importance in various fields of application like food, petroleum industry and pharmacology [4]. During the cooling process, the majority of the droplets remain in a metastable state (i.e. undercooled liquid) and exhibit a stochastic crystallization behaviour [5,6]. Ideally, after rupture of supercooling, the liquid droplets are transformed into solid particles of the same size. Since the droplets in emulsions are commonly polydisperse and the nucleation in emulsions is strongly droplet size dependent, individual and stochastically distributed crystallization often occurs during cooling in oil-in-water dispersions. Heterogeneous crystallization leads to the presence of supercooled liquid and crystallized solid droplets within one dispersion and thus to incomplete crystallization [7,8]. Consequently, supercooled droplets may collide with crystallized particles, wet their surface and crystallize/coalesce with those particles leading to an increase in the particle size distribution [9].

It is interesting to outline that when one liquid is dispersed in into another one, the surface area that separates the two liquids increases considerably, making the emulsion unstable [10,11]. In this case, the coalescence phenomenon becomes a determining factor in the stability of emulsions. This phenomenon will depend on several parameters and especially on temperature. Since droplet crystallization is an exothermic reaction, it can lead to an increase in the temperature of the emulsifying medium and cause a chaotic droplet movement in the emulsion, leading to coalescence. In addition, a high mass fraction of the dispersed liquid increases the contact between the droplets, therefore their coalescence [10].

Any crystallization process starts with nucleation due to the supercooling phenomenon. Usually, nuclei are produced by heterogeneous nucleation mechanism due to the presence of impurities in the liquid droplets [12,13]. Therefore, it is theoretically proved that when heterogeneous nucleation is considered, crystallization takes place at a higher temperature. In water emulsion, the volume of individual droplets is comparatively low and the number of impurities is usually much smaller than the number of droplets [14]. Consequently, nuclei formation in emulsion may approach the homogeneous nucleation mechanism, which requires higher supercooling [10,11] and can be a major barrier to industrial application [15].

Given a relatively small size, crystallization of water droplets in the emulsion is commonly monitored by the rate of nucleation of crystals in the volume. As Turnbull [16] shows this rate depends mainly

* Corresponding author.
E-mail addresses: hamou.elotmany@gmail.com (H. El-Otmany), tarik.elrhafiki@usmba.ac.ma (T. El Rhafiki), tarik.kousksou@univ-pau.fr (T. Kousksou), youssef.zeraouli@univ-pau.fr (Y. Zeraouli).

https://doi.org/10.1016/j.est.2021.102273
Received 28 July 2020; Received in revised form 2 January 2021; Accepted 5 January 2021
Available online 21 January 2021
2352-152X/© 2021 Elsevier Ltd. All rights reserved.

Nomenclature

c_P	Specific heat (J kg^{-1} K^{-1})
J	Probability of crystallization
h	Heat transfer coefficient (W m^2 K^{-1})
k	Thermal conductivity (W m^{-1} K^{-1})
L_F	Latent heat of PCM (J kg^{-1})
N	Total number droplets
$\mathbb{P}$	Mass fraction
q	Heat source
r	Space variable (m)
T	Temperature (K)
t	Time (s)
V	Volume (m^3)
W	Brownian motion

Subscripts, superscripts

E	est
m	Melting
i	Initial
P	Plate temperature
s	Solid
W	West

Greek symbols

β	Cooling rate (C/min^{-1})
ΔS	surface (m^2)
φ	Crystallized fraction mass density

on the size of the droplets and their distribution in the emulsion. In addition, due to the stochastic nature of nucleation, it is not possible to identify where crystallization may occur within the emulsion. However, a probability of the appearance of solid droplets can be estimated because it is possible to estimate the rate of nucleation [16].

Physical models describing the crystallization kinetics of phase change materials (PCMs) such as water are considered important for the thermal analysis of latent thermal energy storage systems in industry [17,18]. Some complex physical phenomena such as supercooling are starting to be incorporated in models [19]. Physical models describing the phase transition with supercooling in PCMs and their validation have recently been presented [20–22]. Uzan et al. [23] outline macroscopic modelling based on the relationship between enthalpy function and temperature in a supercooled liquid and provide some general considerations regarding nucleation theory. Jin et al. [24] present results for the phenomenological modelling of supercooling considering an internal heat source in the PCM which is activated in a defined temperature range. However, no information is given on the selection of the heat source values. The model is experimentally validated for recorded temperature measurements during thermal cycles applied to a PCM building material. This approach was also adopted by Kumarasamy et al. [25]. Davin et al. [26] present a new physical modelling based on the apparent heat capacity to describe the supercooling phenomenon during the crystallization process. Most of the physical models describing the liquid–solid phase transition considering the supercooling phenomenon are based on deterministic approaches [23]. Stochastic modelling of the nucleation process in PCM seems to be limited to the identification of the nucleation probabilities [27,28]. The objective of this work is to present a thermal model to describe a stochastic character of the crystallization of water droplets dispersed in an emulsion.

1.2. Motivation of the study

Modelling of random crystallization of undercooled water droplets dispersed inside an emulsion modelling is challenging due to the combining supercooling effect and simultaneous heat transfer and solidification process. From a numerical point of view, limited work [29–31] has been published to address heat transfer during the crystallization of undercooled water droplets dispersed in an emulsion. Dumas et al. [29] and El Rhafiki et al. [32] presented physical models supercooled water droplets using a deterministic probability law. The numerical results obtained by these authors have been validated by the experimental data. Unfortunately, the models developed by these authors cannot detect the local heating of concentrated emulsions during the water droplet crystallization process. This local heating phenomenon has been experimentally observed by Krichi [33] for concentrated emulsions. Vallet and Trujillo [34] reported a mathematical model to describe the local heating within the emulsion due to the stochastic crystallization of water droplets. The numerical results of this model have not been compared with the experimental results obtained by Krichi [33]. Therefore, the objective of this work is to introduce a new stochastic mathematical model, inspired by the Black–Scholes model in finance, to describe the random crystallization of undercooled water droplets scattered inside an emulsion.

The paper is organized as follows: we present our stochastic mathematical model, inspired by the Black–Scholes model in finance, for describing the crystallization of water droplets dispersed in an emulsion. The energy equation was adopted to analyse the heat transfer during the crystallization of water droplets in the emulsion. Next, the Finite Volume Method (FVM) combined with the tri-diagonal Matrix Algorithm (TDMA) to solve iteratively the direct model is deployed. The inverse method is then used to estimate the unknown parameters for stochastic model by using the experimental temperature measured on the axis of cylinder. Finally, we present some numerical results to verify our approach and we give some conclusions.

2. Stochastic and physical models

In this section, we introduce and elaborate our stochastic crystallization model used throughout the remainder of this paper. We also provide the heat transfer model within the emulsion, the discretization method and the genetic algorithm to identify the unknown parameters in the stochastic model (see Section 3.2).

2.1. Stochastic model of crystallization

In emulsion, the crystallization inside each droplet occurs over two different phases: the nucleation process and the crystalline growth, triggered by a supercritical germ, leading to the total solidification of the droplet [6]. In this study, we will mainly focussed on water droplet crystallization dispersed in an emulsion. Due to the relatively small droplet dimensions, we can assume that the crystallization of dispersed water droplets is driven only by the rate of nucleation of crystals within the droplet volume [5]. It is important to point out that the steady cooling of a water emulsion presents the occurrence of supercooling (the persistence of the liquid state of the water below its melting temperature T_m) as well as the unpredictable nature of water crystallization in droplets [31]. This means that water droplets do not simultaneously crystallize at the same temperature and crystallization inside emulsion takes place over a range of temperatures [6,7]. Due to the stochastic nature of nucleation, it is not possible to determine the location where crystallization began inside the emulsion. However, a probability of the occurrence of solid droplets can be estimated as it is possible to estimate the nucleation rate [8].

Assuming a population of droplets N (of a quasi-mono-disperse emulsion) with volumes V that are cooled at a constant cooling rate. Let $n(r,t)$ be a proportion of N which corresponds to the number

of solidified droplets in a control volume at the position r and at the time t. Crystal growth in each control volume is assumed to be independent of other control volumes and is controlled by the identical deterministic model. To take into account the crystal growth fluctuations in control volume and physical phenomena not captured by the deterministic term, a stochastic component is added as part of the nucleation probability calculation. In [34], Vallet and Trujillo proposed a stochastic model by adding an implicit random part of the nucleation rate to deterministic contribution of crystallized droplets, under some hypothesis, such that

$$\frac{\partial \varphi(r,t)}{\partial t} = (1-\varphi(r,t))J(T) + b(T,\varphi(r,t))\frac{\partial W_t}{\partial t}, \tag{1}$$

where

1. φ is the crystallized fraction of droplets such that

$$\varphi := \varphi(r,t) = \frac{n(r,t)}{N} \tag{2}$$

2. $(W_t)_{t\geq 0}$ is called the Wiener process or standard Brownian motion in a one-dimension space. It is a continuous stochastic process with stationary independent increments and its variance proportional to the time increment (see [35,36]) such that

$$W_0 = 0, \quad W_t - W_s \sim \mathcal{N}(0, t-s) \quad \forall 0 \leq s \leq t \tag{3}$$

where $\mathcal{N}$ indicate the normal distribution with mean 0 and variance $(t-s)$.

3. $b(T,\varphi)$ is the undefined function of the temperature and the nucleation rate and it satisfies the following hypothesis:

   H1. $b(T,\varphi)$ is Lipschitz continuous and positive function on $\mathbb{R}^2$.
   H2. $b(T,\varphi)$ vanishes in outside of the crystallization area. It means that if $T \geq T_m$ and $T \leq 0$ for all values of φ and if $\varphi \leq 0$ and $\varphi \geq 1$ for all values of T.

4. $J(T)$ is the deterministic nucleation rate.

Moreover, under $H1 - H2$ and some hypothesis on J, the authors proved the existence and the uniqueness of the solution to stochastic model based on the theory of nonlinear stochastic equations (for complete presentation, we refer to [34,37,38]). They presented also the numerical tests, but they did not compare their results against an experimental model and they defined explicitly the coefficient part to correct the number of crystallized droplets; b. To tackle the previously mentioned problem, we propose here to modify a stochastic model by using new function associated to Brownian motion based on Black–Scholes model (see [39]) with deterministic proportion (drift) and diffusion independently to the nucleation rate and the probability of crystallization J. Thus, we propose the stochastic model for the speed crystallization in dispersed media:

$$\frac{\partial \varphi}{\partial t} = (1-\varphi)J(T) + (1-\varphi)S(T)\frac{\partial W_t}{\partial t}, \tag{4}$$

with $S(T)$ the diffusion associated to one-dimensional Brownian motion W_t and its expression will be discussed. A stochastic model (4) can be written in a more concise form by using the generic stochastic differential equation:

$$d\varphi = (1-\varphi)J(T)dt + (1-\varphi)S(T)dW_t. \tag{5}$$

Remark 2.1. The last hypothesis $H2$ is not essential for determining a solution of the mathematical problem and to reproduce the numerical results presented in [34]. Indeed, in the case of $H2$, one should replace the part stochastic b by another function such that $S(T_m) = \alpha_0 \neq 0$.

In the sequel and for the sake of simplicity, we denote $J_d(T)$ the deterministic part of the nucleation rate and $J_s(T)$ the stochastic part of the mathematical model. According to nucleation theories [9], the probability of crystallization can be determined from the following expression:

$$J_d(T) := J(T) = \begin{cases} A\exp\left(-\dfrac{B}{T(T-T_m)^2}\right) & if \quad T < T_m, \\ 0 & otherwise \end{cases} \tag{6}$$

where A and B are assumed to be constant for a given volume of the water droplet [8,31] and $\Delta T := T - T_m$ is the degree of supercooling. A detailed description of the method to determine $J_d(T)$ may be found in [8,31].

According to the stochastic mathematical model in Eq. (4), we consider

$$J_s(T) = S(T)\frac{\partial W_t}{\partial t} \tag{7}$$

and the part of diffusion $S(T)$ is expressed in the form:

$$S(T) = \begin{cases} \alpha_0 + \alpha_1\left(T_m - T\right)^3 & if \quad T < T_m, \\ \alpha_0 & otherwise \end{cases}$$

where α_0 and α_1 ((°C)$^{-3}$) are the positive real numbers. A judicious choice of the best parameters values involved in the function S, which ensure and fit the experimental data of the crystallization model, is defined by using optimization techniques. For this study, a Genetic Algorithm optimization [30] is selected and used for this nonlinear mathematical model to estimate the both parameters from the experimental data, see Section 3.2. Because it does not need to compute the derivative of the misfit function which is a benefit advantage for our problem whose misfit function is not known in analytic expression. Additionally, it is used for stochastic differential equation to estimate some parameters (see [40]) and it avoids the subjectivity and some ill-posed problem associated with standard optimization methods.

Thus, as stated by using the previous notations, the mathematical model in Eq. (4) become

$$\frac{\partial \varphi}{\partial t} = (1-\varphi)\left(J_d(T) + J_s(T)\right). \tag{8}$$

2.2. Physical model

In order to validate our physical model, the experimental protocol developed by Krichi [33] was used. It consists of a metallic cylinder (see Fig. 1) filled with a water emulsion. The cylinder volume is about 500 cm^3 and two isolated plugs close it. A detailed description of the experimental device may be found in [3,4,31]. A number of thermocouples are placed in the cylinder and are uniformly positioned according to the radius and the height of the cylinder. The cylinder is immersed within a tank filling with a liquid solution at temperature T_b, which can be heated or cooled at a constant rate β using a thermostat. The height of cell is sufficiently larger than its diameter to consider that only radius r and time t are the variables of the study. The experimental conditions are accurately monitored to crystallize only the water droplets, which are supposed to be homogeneously distributed in the emulsion [3,4]. We assume that the thermophysical properties remain constant during the cooling process. However, these properties vary according to the dispersed phase being in solid, liquid or in an intermediate state (liquid–solid). The thermophysical properties of the emulsion are dependent on the mass fraction $\mathbb{P}$ of the water droplets dispersed in the emulsion [3].

Two effects drive the heat transfer inside the emulsion:

- The steady cooling process monitored by the temperature T_b.
- The stochastic crystallization of the water droplets inside the emulsion.

In our case, the effect of heat transfer by radiation has been neglected since crystallization process takes place at very low temperatures. The droplets are assumed to be immobile and homogeneously dispersed inside the emulsion. Therefore, there is no flow in the emulsion

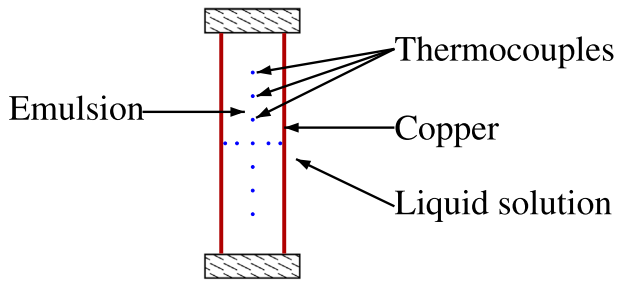


Fig. 1. Experimental device.

and the natural convection in the droplets during the crystallization process has been neglected.

Following the previous assumptions, the heat transfer during the cooling process of the emulsion can be described using the energy equation:

$$\rho c_p \frac{\partial T(r,t)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rk\frac{\partial T(r,t)}{\partial r}\right) + \dot{q}(r,t), \quad 0 < r < R, \quad \forall t > 0 \tag{9}$$

where k is the thermal conductivity of the emulsion, ρ is the mass density and c_p is the specific heat capacity.

The heat source $\dot{q}$ describe the crystallization process inside the emulsion. This term is proportional to the number of droplets crystallizing per units of time and volume of emulsion

$$\dot{q}(r,t) = \rho_0 \mathbb{P} L_F \frac{\partial \varphi}{\partial t}, \quad 0 < r < R, \quad \forall t > 0 \tag{10}$$

where ρ_0, $\mathbb{P}$, L_F indicate respectively the mass density of water, the mass fraction of water and the latent heat of fusion.

Taking into account Eqs. (8) and (10), Eq. (9) of the heat transfer becomes:

$$\rho c_p \frac{\partial T(r,t)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rk\frac{\partial T(r,t)}{\partial r}\right) + \rho_0 \mathbb{P} L_F \left(1-\varphi(r,t)\right)\left(J_d(T) + J_s(T)\right), \quad 0 < r < R, \quad \forall t > 0. \tag{11}$$

According to the experimental device, the boundary conditions complete Eq. (11):

$$\left.\frac{\partial T(r,t)}{\partial r}\right|_{r=0} = 0 \text{ and } -k\left.\frac{\partial T(r,t)}{\partial r}\right|_{r=R} = h_{ext}\left(T(R,t) - T_b(t)\right) \tag{12}$$

where h_{ext} is the convective heat transfer coefficient between the bath and the emulsion. As for initial conditions, we assume: $T(r,0) = T_0$, $\forall r > 0$. The bath temperature $T_b(t)$ is programmed according to the following formula:

$$T_b(t) = \beta t + T_0, \quad \beta < 0 \tag{13}$$

where β is the cooling rate and T_0 is the bath temperature at $t = 0$. During the cooling process, $T_b(t)$ decreases linearly until reaching T_s. Then, $T_b(t)$ is maintained at T_s.

In the next section we will give an alternative method for solving the direct physical model using the finite volume method (FVM) and we will identify the parameters involves in stochastic model by using Genetic algorithm optimization.

3. Methodology for discretization and parameter identification

The present section is divided in two subsection. The first one consists in employing the finite volume method (FVM) to solve heat transfer equation (11) and using the Box–Muller method [41] to simulate a Brownian motion described in (8). In the second subsection, we present briefly a genetic algorithm (GA) in order to identify the parameters α_0 and α_1 described in (8) by performing a technique both experimental and numerical temperatures.

3.1. Finite volume method for solving direct physical model

The strategy of finite volume method is to regard the computation domain as subdivided into a collection of finite volumes, see for example [42] and references therein. This method involves three basic steps: we first generate the grid by dividing the domain into small discrete control volume and we place the numbers of nodal points in the given domain where each node is surrounded by a control volume or cell, and the physical boundaries coincide with the control volume boundaries. Then, we integrate over the control volume to yield a discretized equation at its nodal point. Finally, we use Tri-diagonal Matrix Algorithm (TDMA), also as known under the name of Thomas algorithm to solve iteratively the set of discretized equation. But before doing so, we need to introduce an appropriate notations and some considerations.

Let P be a general nodal point of control volume with two neighbours nodes, in one dimension geometry, on the east and west are noted as follow; E and W respectively. We denote by 'e' and 'w', the two faces of the volume control at their sides respectively.

We assume an uniform grid size of the control volumes and we use a full implicit scheme in time with constant time steps Δt.

By integrating Eq. (11) over the span of each control volume at node P and using (10), we obtain

$$\int_t^{t+\Delta t}\int_V \rho c_p \frac{\partial T(r,s)}{\partial t}\, dV ds = \int_t^{t+\Delta t}\int_V \frac{1}{r}\frac{\partial}{\partial r}\left(rk\frac{\partial T(r,s)}{\partial r}\right) dV ds + \int_t^{t+\Delta t}\int_V \rho_0 \mathbb{P} L_F \frac{\partial \varphi}{\partial t}\, dV ds. \tag{14}$$

By putting together (4), $dV = rdr$ and by integrating between 'w' and 'e', we can get:

$$\begin{aligned}\rho c_p \left(T_P^{t+\Delta t} - T_P^t\right) r_P \Delta r = &\left(rk\frac{\partial T(r,s)}{\partial r}\bigg|_e - rk\frac{\partial T(r,s)}{\partial r}\bigg|_w\right)\Delta t \\ &+ \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) J(T_P) r_P \Delta r \Delta t \\ &+ \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) S(T_P)\left(W_{t+\Delta t} - W_t\right) r_P \Delta r.\end{aligned} \tag{15}$$

By discretizing the first-order derivative by a second-order central difference scheme, Eq. (15) can be written as follow:

$$\begin{aligned}\rho c_p \left(T_P^{t+\Delta t} - T_P^t\right) r_P \Delta r = &\left(r_e k\frac{T_E^{t+\Delta t} - T_P^{t+\Delta t}}{\delta r_e} - r_w k\frac{T_P^{t+\Delta t} - T_W^{t+\Delta t}}{\delta r_w}\right)\Delta t \\ &+ \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) J(T_P^{t+\Delta t}) r_P \Delta r \Delta t \\ &+ \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) S(T_P^{t+\Delta t})\left(W_{t+\Delta t} - W_t\right) r_P \Delta r.\end{aligned} \tag{16}$$

We can then solve for $T_P^{t+\Delta t}$ by dividing by Δt, and collecting like terms in Eq. (16):

$$a_P T_P^{t+\Delta t} = a_W T_W^{t+\Delta t} + a_E T_E^{t+\Delta t} + a_P^t T_P^t + S_{deter}^{t+\Delta t} + S_{stoch}^{t+\Delta t} \tag{17}$$

where

$$\begin{aligned}&a_P^t = \rho c_p r_P \frac{\Delta r}{\Delta t}, \quad a_W = k\frac{r_w}{\delta r_w}, \quad a_E = k\frac{r_e}{\delta r_e}, \quad a_P = a_W + a_E + a_P^t, \\ &S_{deter}^{t+\Delta t} = \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) J(T_P^{t+\Delta t}) r_P \Delta r, \\ &S_{stoch}^{t+\Delta t} = \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) S(T_P^{t+\Delta t}) r_P \frac{\Delta r}{\Delta t}\left(W_{t+\Delta t} - W_t\right),\end{aligned} \tag{18}$$

J is the nucleation probability and S is the function used to modulate the importance of Brownian motion in physical model. It involves two parameters (α_0, α_1) which we will be estimated in Section 3.2.

For the nodes adjacent to borders, they have a face value given by replacing the border conditions (12) in (17).

We now interest in the stochastic term $S_{stoch}^{t+\Delta t}$ containing the Brownian motion. By putting together (3) and $\mathcal{N}(0,\Delta t) \sim \sqrt{\Delta t}\mathcal{N}(0,1)$, $S_{stoch}^{t+\Delta t}$ can be expressed in law as follow:

$$S_{stoch}^{t+\Delta t} \sim \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) S(T_P^{t+\Delta t}) r_P \frac{\Delta r}{\sqrt{\Delta t}}\mathcal{N}(0,1) \tag{19}$$

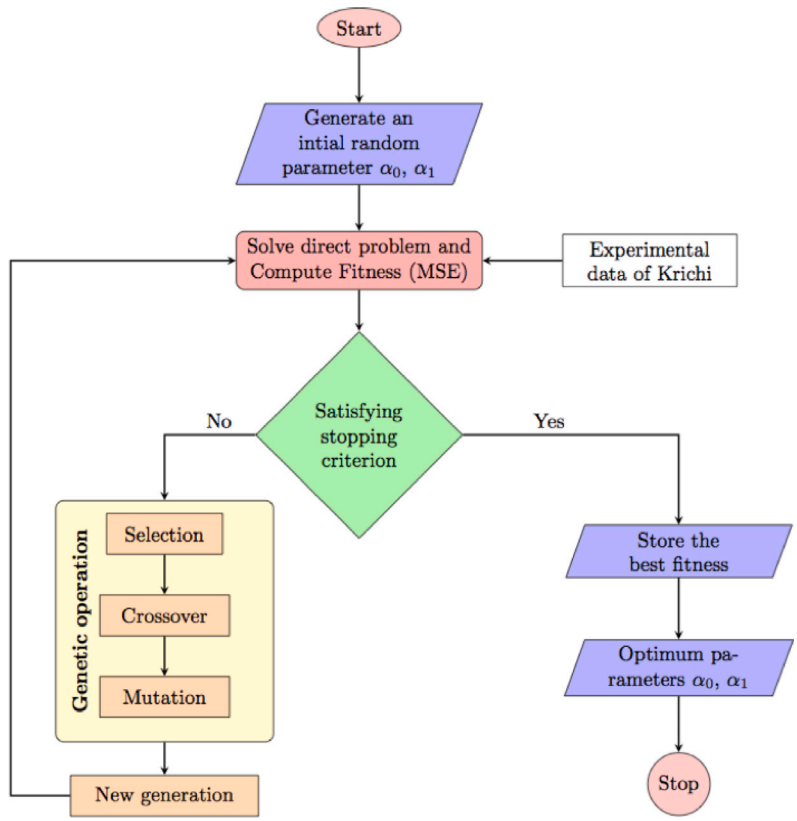


Fig. 2. Flowchart of the used methodology.

where $\mathcal{N}(0,1)$ is a standard normal random with mean 0 and variance 1. Then, by putting $Z \sim \mathcal{N}(0,1)$, we can write

$$S_{stoch}^{t+\Delta t} \sim \rho_0 \mathbb{P} L_F \left(1-\varphi_P\right) S(T_P^{t+\Delta t}) r_P \frac{\Delta r}{\sqrt{\Delta t}} Z \tag{20}$$

In order to simulate a standard normal random Z, we adopt a Box–Muller method [41] based on uniform distribution, $\mathcal{U}(0,1)$. Because, it uses the logarithmic function and trigonometric computational, whereas algorithms based on the inverse transformation usually use simple polynomial evaluations. Moreover, its is really simple in term of programming and this is why I think it is a really good one.

The Box–Muller method generates a pair of independent standard normal random variables using a pair of uniform random numbers as follows:

- U_1 and U_2 be two independent variables uniformly distributed on $(0,1)$;
- Compute $Z_1 = \sqrt{-\log(U_1)}\cos(2\pi U_2)$, $Z_2 = \sqrt{-\log(U_1)}\sin(2\pi U_2)$.

By concatenating Z_1 and Z_2, we finally simulate the standard random variable Z.

To finish out the finite volume method, we use TDMA method to solve iteratively the above discretized equations.

3.2. Genetic algorithms for solving inverse problem

In the direct physical model defined by a set of partial differential equations in Eqs. (9), (8) and (12). Once the geometrical properties and all the physical parameters (ρ, ρ_0, c_p, k, $\mathbb{P}$, L_F) are known and reported in [27]. For the inverse model, it is assumed that the parameters (A, B) of nucleation probability J and the parameters (α_0, α_1) of the stochastic model given by (4) are unknown.

The objective of the inverse problem is to determine the unknown coefficients by using experimental temperature measured on the axis of the cylinder ($r = 0$), see [33] for more details.

For the sake of the present study, we assume that the kinetics parameters A and B of probability J are fixed and reported in [28]. So, the estimation of (α_0, α_1) from measured can be constructed as an optimum control problem which aims at minimizing the least square norm $\Psi(\mathbf{m})$:

$$\Psi(\mathbf{m}) = \sum_{i=1}^{N} \left(\frac{T^{exp}(t_i) - \mathbf{E}(T^{num}(\mathbf{m}, t_i))}{\sigma_i} \right)^2 \tag{21}$$

where $\mathbf{m} = (\alpha_0, \alpha_1)$ is the set of unknown parameters, N is the total number of measurements, $T^{exp}(t_i)$ are the experimental temperatures, $\mathbf{E}(T^{num}(\mathbf{m}, t_i))$ are the expected temperatures generated from the computed solution of the direct physical model (Section 3.1) and σ_i are the variance of the measurement errors. The solution of the inverse problem is obtained by minimizing the least square norm $\Psi(\mathbf{m})$, (21).

The Genetic Algorithm (GA) was adopted for solving the inverse problem, for more details see [43–46] and references therein. The GA uses a set of chromosomes referred to as a population and its uses also a binary encoding of solutions, which means that the potential solution of the raised problem is represented by a binary code (chromosome). Moreover, it involves five essentials components: fitness function, population, selection, crossover and random mutation such that the crossover component is based on the so-called Roulette Wheel Selection, see [47–49] for more details.

The GA computational procedure for the inverse problem is summarized in Fig. 2 and was developed under FORTRAN language.

Table 1 reports initial setting parameters used in GA for this study. In addition, the population size of GA is 30 and the algorithm is accomplished by 450 iterations. It should be mentioned that for each parameter an initial guess by using uniform distribution between 0 and 1 is made, and according to the parameter ranges, the initial random

Table 1
The specification of the Genetic algorithm.

GA's parameter	Parameter value
Mutation percentage	35
Mutation rate	0.03
Crossover percentage	75
Crossover adjustment	0.05

Table 2
Prescribed range of the parameters (α_0, α_1) given to GA for evaluating the misfit function of direct problem and their obtained optimum value.

Unknown parameter	Proposed range	Optimum value
α_0	[0.01, 50]	37.37654 ± 0.01
α_1	[278, 325]	309.71835 ± 0.001

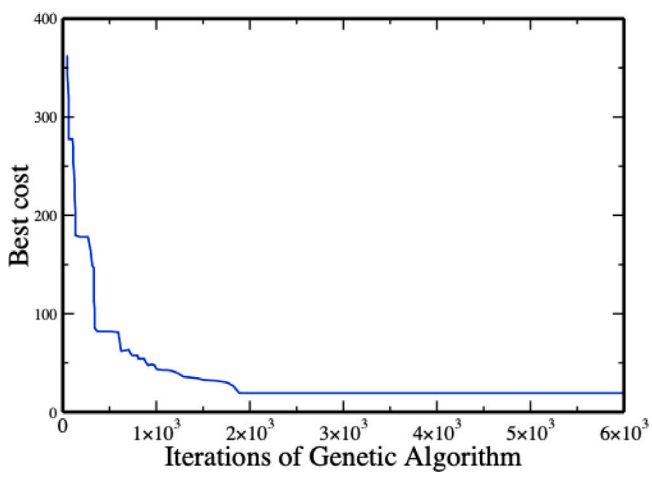


Fig. 3. Misfit function value of the Genetic algorithm in the inverse problem.

is transformed to its corresponding value. Eventually, the output of the model is obtained based on the mapped initial guess of the parameters (α_0, α_1) and it will be evaluated by the least square norm Ψ, (21).

By accomplishing the GA computational procedure of the inverse model based on the experimental temperatures of Krichi [33], the optimum value of the unknown parameters (α_0, α_1) of the direct model is obtained. Fig. 3 illustrates the value of the misfit function among all 30 populations with respect to the Number of misfit Function Evaluation.

Table 2 reports the range of the parameters (α_0, α_1) of the direct physical model and their optimum values obtained. It should mentioned that the range data regarding the intercept parameter α_0 is a constant having no units and the rang data regarding the parameter of perturbation α_1 is in $(°C)^{-3}$. It reveals that the optimum parameters (α_0, α_1), are peaked along the corresponding ranges and are slightly far from the limits. This can be considered as an indication for which the proposed ranges of two parameters (α_0, α_1) are well-defined.

By using the previous optimum values, the output of the direct physical model are computed. Fig. 4 shows the physical model output in comparison to the experimental temperatures. The foregoing figure reveals that although the proposed model is successful to get the appropriate result, it fails to provide a fitting data.

4. Numerical simulations and discussions

In this section, we present some numerical simulations in order to validate the Brownian motion model and to illustrate the effect of water mass concentration on crystallization kinetics within emulsion. The thermophysical properties and test-case are taken from [33]. Let $\mathbb{P} = 0.50$, $T_m = 0\,°C$ and $T_s = -68\,°C$. The numerical simulations are carried out with the FORTRAN language developed by our team.

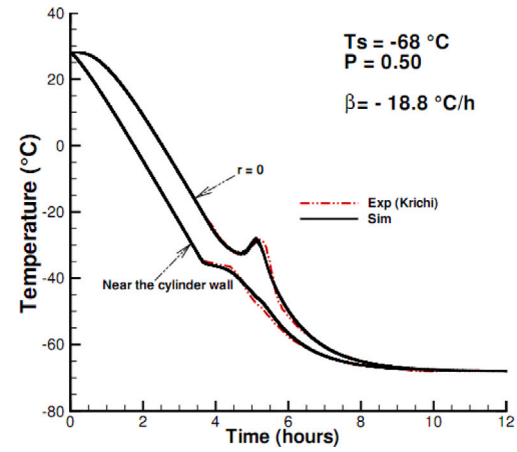


Fig. 4. Experimental and numerical results.

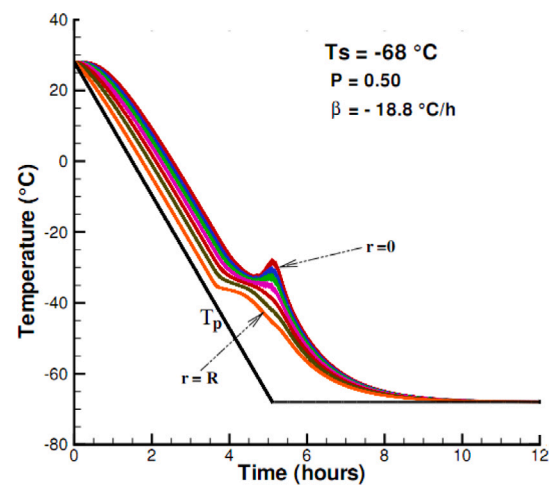


Fig. 5. Emulsion temperature versus time at different positions inside the cylinder.

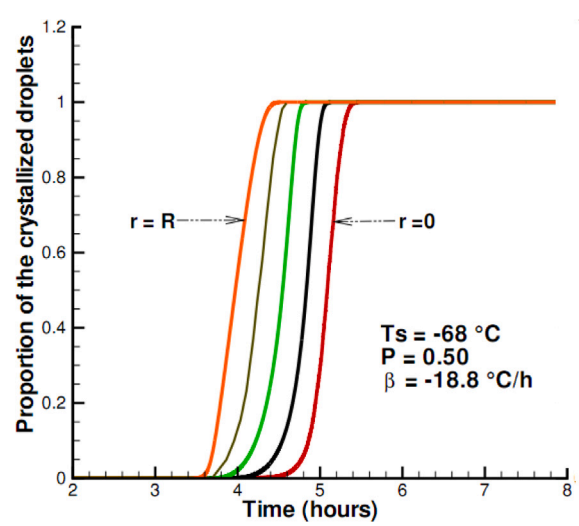


Fig. 6. Proportion of the crystallized droplets versus time.

Fig. 4 compares the experimental (dashed line) [33] and numerical results (solid line). This figure shows the variation of the emulsion temperature at the center ($r = 0$) and extremity ($r = R$) of the cylinder versus time for $\beta = -18.8\,°C/h$ and $T_s = -68\,°C$. A good

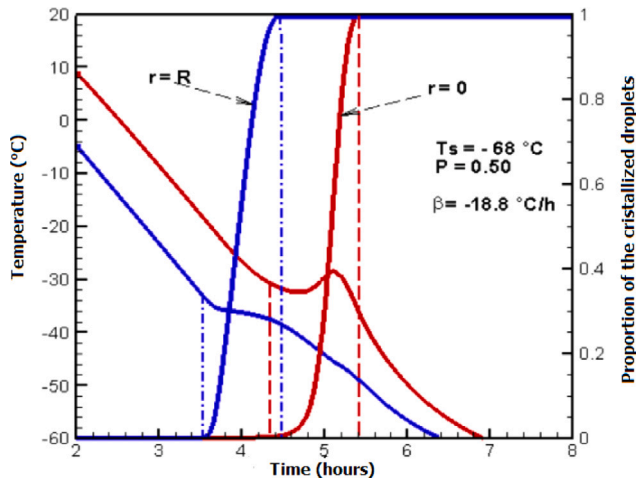


Fig. 7. Emulsion temperature and proportion of the crystallized droplets versus time.

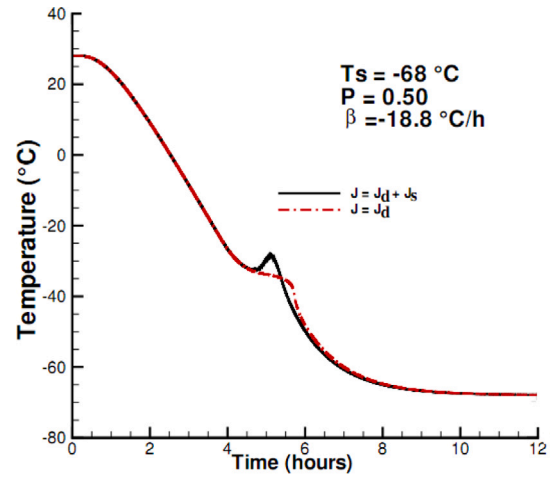


Fig. 9. Comparison between deterministic and stochastic model.

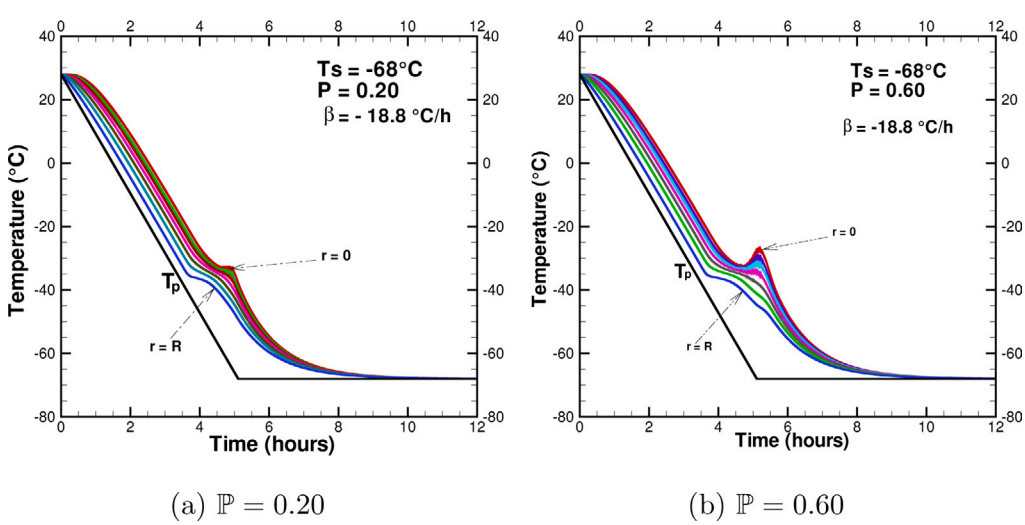
concordance can be observed between the experimental and numerical results. As intended, the water droplets near the inner face of the cylinder begin to crystallize first at around the temperature of −38 °C. The thermal energy released during the solidification of these droplets heats the central region of the cylinder, causing a significant delay in the water droplets crystallization in this zone. One can also remark that the droplet crystallization in the center of the cylinder has caused an important increase in the emulsion temperature. Such temperature increase can be explained by a rapid crystallization of a large quantity of water droplets, which can cause a chaotic droplet movement in the emulsion, leading to coalescence phenomenon.

At the beginning of the cooling process, all points inside the cylinder have the same temperature (see Fig. 5). As soon as the cooling process is activated, the temperature at each point decreases linearly until the crystallization temperature is reached. It can be seen that the freezing process takes place at practically around −38 °C. We can also note that the degree of supercooling in this case is very important. After the end of crystallization, the emulsion temperature decreases linearly to reach the final stabilization temperature that we have programmed.

Further information on the kinetics of phase changes within the emulsion can be obtained by presenting the evolution of the proportion of crystallized droplets in the emulsion versus time (see Fig. 6). Fig. 6 shows that the crystallization front within the emulsion moves from $r = R$ towards the center of the cylinder $r = 0$. As illustrated in Fig. 7, the percentage of crystallized droplets remains low at the beginning of the emulsion temperature stabilization around 38 °C. For $r = R$, the percentage of crystallized droplets increases significantly only at the end of the temperature plateau. For $r = 0$, the percentage of crystallized droplets remains practically negligible during the stagnation of the emulsion temperature at around 38 °C. Then, the progressive crystallization of the droplets leads to the release of heat, which will contribute to the increase of the emulsion temperature in this region.

Fig. 8 describes the effect of the droplet concentration $\mathbb{P}$ on the kinetics of the droplet crystallization inside the emulsion. For $\mathbb{P} = 0.6$, the progressive and stochastic crystallization of a large number of droplets in the center of the cylinder provokes a release of heat which leads to an increase in the emulsion temperature in this zone. We can also notice that for low concentrations of droplets (i.e. $\mathbb{P} = 0.2$) the crystallization takes place at almost constant temperature. It is important to remember that the contact surface between droplets increases considerably with increasing $\mathbb{P}$, making the emulsion unstable and enhancing the possibility of droplet coalescence. The obtained results confirm that for low values of $\mathbb{P}$, the risk of coalescence of the droplets is negligible and the droplets crystallization occurs gradually and deterministically.

Fig. 9 compares the results of the two crystallization models: deterministic and stochastic. It can be remarked that if the phase change process in the emulsion is described only by a deterministic law of



(a) $\mathbb{P} = 0.20$ (b) $\mathbb{P} = 0.60$

Fig. 8. Emulsion temperature versus time (a) $\mathbb{P} = 0.20$ and (b) $\mathbb{P} = 0.60$.

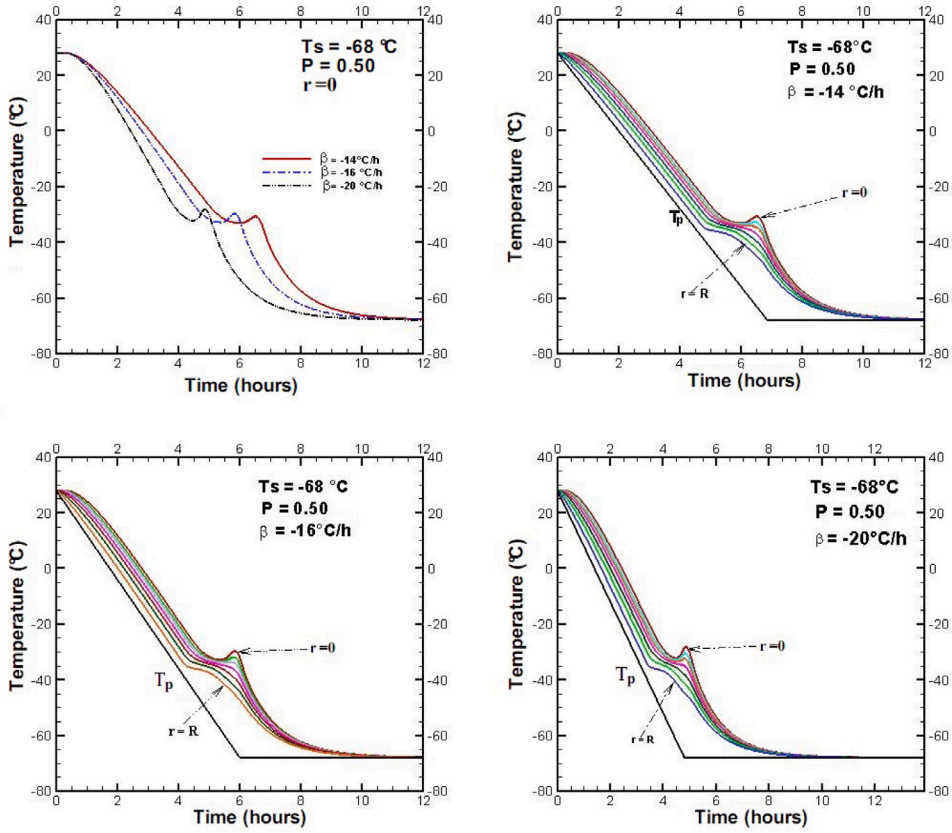


Fig. 10. Effect of cooling rate β on crystallization kinetics in the emulsion.

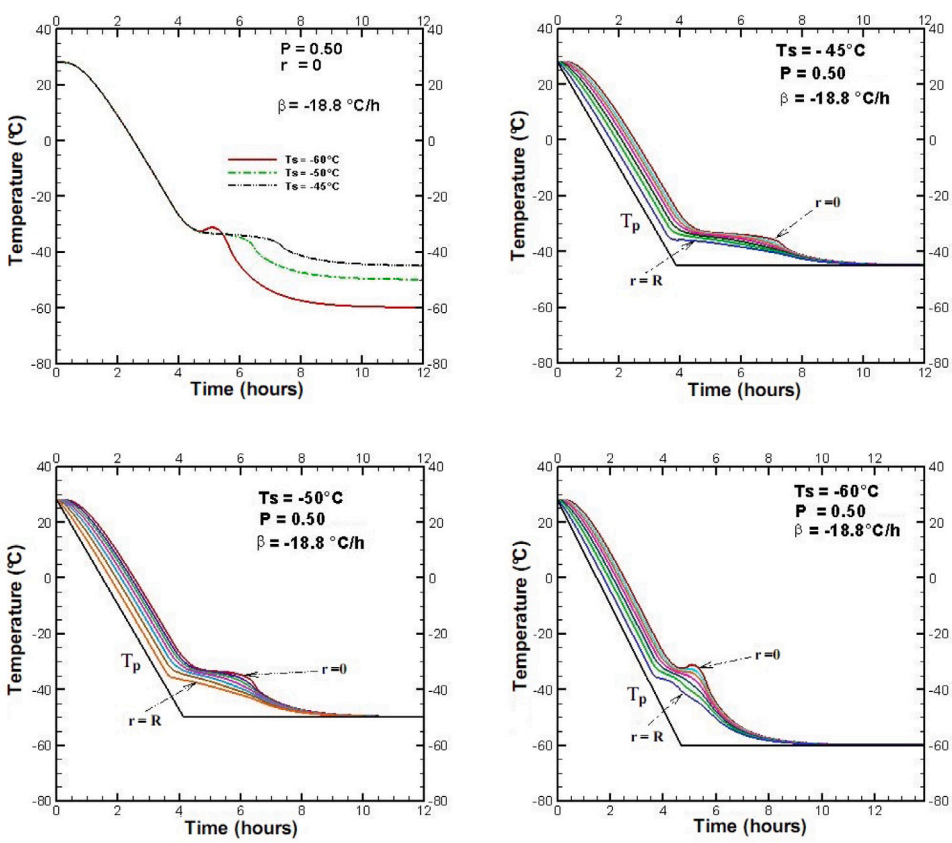


Fig. 11. Effect of the stabilization temperature T_s on the crystallization kinetics inside the emulsion.

probability J_d, the droplets crystallize progressively over a range of temperature. However, in our case, the deterministic crystallization of the droplets is unable to detect the temperature increase inside the emulsion. Nevertheless, if the kinetics of phase transition inside the emulsion is described using stochastic law of probability ($J_d + J_s$), the crystallization of the dispersed phase provokes a rise in temperature inside the emulsion. We think that stochastic crystallization of the dispersed phase initiates a random displacement of this phase inside the emulsion, which can provoke coalescence of water droplets and render that emulsion unstable.

Fig. 10 shows the effect of the cooling rate on the kinetics of crystallization inside the emulsion. Three cooling rates were tested: −14 °C/h, −16 °C/h and −20 °C/h. It can be noticed that the crystallization time of the droplets in the emulsion decreases with increasing cooling rate. We can also observe that the increase in the cooling rate had no impact on the stochastic character of the crystallization.

Fig. 11 presents the impact of the stabilization temperature T_s on the kinetics of crystallization inside the emulsion. Three stabilization temperatures T_s were tested: −45 °C, −50 °C and −60 °C. We notice that the crystallization of the water droplets in the center of the cylinder takes place at practically constant temperature for both stabilization temperatures: −45 °C and −50 °C. We can also notice that when the temperature T_s approaches the crystallization temperature of the water droplets, the duration of the phase change in the emulsion increases. We can conclude that the kinetics of phase change in emulsions depends significantly on the choice of temperature T_s.

5. Conclusion

The main objective of this paper consisted in proposing a close-to-reality a stochastic model to describe a heat transfer coupled with a random crystallization of undercooled water dispersed within emulsions. As the first step, a Finite Volume method is used to solve the direct physical model. This method is based on a full implicit scheme, to discretize a heat transfer equation, and, a Box–Muller method, with unknown parameters, to simulate a Brownian motion. In order to compensate the difference of the output physical model and the experimental temperature obtained from literature, the inverse problem is solved by using Genetic Algorithm. All simulations results demonstrated that the proposed model is accurate and efficient when compared with the experimental temperature. Future works are intended to be done in this field, such as the use of neural network/Deep learning in order to identify all parameters and fit quickly the measured temperature.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors of the paper would like to thank the three anonymous reviewers for their insightful suggestions and careful reading of the manuscript.

References

[1] D.J. McClements, Crystals and crystallization in oil-in-water emulsions: implications for emulsion-based delivery systems, Adv. Colloid Interface Sci. 174 (2012) 1–30.
[2] K. Köhler, A. Hensel, M. Kraut, H.P. Schuchmann, Melt emulsification is there a chance to produce particles without additives? Particuology 9 (5) (2011) 506–509.
[3] K. Köhler, H.P. Schuchmann (Eds.), Emulgiertechnik: Grundlagen, Verfahren und Anwendungen, third ed. Behr, Hamburg, 2012.
[4] S. Abramov, P. Ruppik, H. Schuchmann, Crystallization in emulsions: a thermo-optical method to determine single crystallization events in droplet clusters, Processes 4 (3) (2016) 25.
[5] S. Abramov, A. Berndt, K. Georgieva, P. Ruppik, H.P. Schuchmann, Investigation of the infuence of mean droplet size and shear rate on crystallization behavior of hexadecane-in-water dispersions, Colloids Surf. A 529 (2017) 513–522.
[6] D. Clausse, F. Gomez, I. Pezron, L. Komunjer, Dalmazone, Morphology characterization of emulsions by differential scanning calorimetry, Adv. Colloid Interface Sci. 117 (2005) 59–74.
[7] J. Cheng, O.E. Dudu, D. Wang, X. Li, T. Yan, Influence of interfacial adsorption of glyceryl monostearate and proteins on fat crystallization behavior and stability of whipped-frozen emulsions, Food Chem. 310 (2020) 125949.
[8] D. Clausse, J.P. Dumas, Supercooling, Crystallization and Melting Within Emulsion and Divided Systems: Mass, Heat Transfers and Stability, Bentham eBooks, 2016.
[9] D. Kashchiev, N. Kaneko, K. Sato, Kinetics of crystallization in polydisperse emulsions, J. Colloid Interface Sci. 208 (1998) 167–177.
[10] L. Goibier, S. Lecomte, F. Leal-Calderon, C. Faure, The effect of surfactant crystallization on partial coalescence in o/w emulsions, J. Colloid Interface Sci. 500 (2017) 304–314.
[11] J. Bibette, F. Leal-Calderon, Surfactant-stabilized emulsions, Curr. Opin. Colloid Interface Sci. 1 (1996) 746–751.
[12] C. Himawan, V.M. Starov, A.G.F. Stapley, Thermodynamic and kinetic aspects of fat crystallization, Adv. Colloid Interface Sci. 122 (1–3) (2006) 3–33.
[13] D. Clausse, Thermal behaviour of emulsions studied by differential scanning calorimetry, J. Therm. Anal. 51 (1998) 191–201.
[14] E. Günther, T. Schmid, H. Mehling, S. Hiebler, L. Huang, Subcooling in hexadecane emulsions, Int. J. Refrig. 33 (8) (2010) 1605–161.
[15] L. Royon, G. Guiffant, Heat transfer in paraffin oil/water emulsion involving supercooling phenomenon, Energy Convers. Manage. 42 (2001) 2155–2161.
[16] D. Turnbull, Kinetics of solidification of supercooled liquid mercury droplets, J. Chem. Phys. 20 (1952) 411.
[17] Y. Zhao, X. Zhang, X. Xu, S. Zhang, Research progress in nucleation and supercooling induced by phase change materials, J. Energy Storage 27 (2020) 101156.
[18] M.H. Zahir, S.A. Mohamed, R. Saidur, F.A. Al.Sulaiman, Supercooling of phase change materials and the techniques used to mitigate the phenomenon, Appl. Energy 240 (2019) 793–817.
[19] R. Waser, S. Maranda, A. Stamatiou, M. Zaglio, J. Worlitschek, Modeling of solidification including supercooling effects in a fin-tube heat exchanger based latent heat storage, Sol. Energy 200 (2020) 10–21.
[20] L. Huang, E. Günther, C. Döetsch, H. Mehling, Subcooling in PCM emulsions - part 1: experimental, Thermochim. Acta 509 (1) (2010) 93–99.
[21] T. Barz, A. Sommer, Modeling hysteresis in the phase transition of industrial-grade solid/liquid PCM for thermal energy storages, Int. J. Heat Mass Transfer 127 (2018) 701–713.
[22] E. Günther, L. Huang, H.Mehling, C. Dötsch, Subcooling in PCM emulsions–Part 2: Interpretation in terms of nucleation theory, Thermochim. Acta 522 (1) (2011) 199–204.
[23] A.Y. Uzan, Y. Kozak, Y. Korin, I. Harary, H. Mehling, G. Ziskind, A novel multidimensional model for solidification process with supercooling, Int. J. Heat Mass Transfer 106 (2017) 91–102.
[24] X. Jin, H. Hu, X. Shi, X. Zhou, L. Yang, Y. Yin, X. Zhang, A new heat transfer model of phase change material based on energy asymmetry, Appl. Energy 212 (2018) 1409–1416.
[25] K. Kumarasamy, J. An, J. Yang, E.-H. Yang, Novel CFD-based numerical schemes for conduction dominant encapsulated phase change materials (EPCM) with temperature hysteresis for thermal energy storage applications, Energy 132 (2017) 31–40.
[26] T. Davin, B. Lefez, A. Guillet, Supercooling of phase change: A new modeling formulation using apparent specific heat capacity, Int. J. Therm. Sci. 147 (2020) 106121.
[27] S. Gibout, A. Jamil, T. Kousksou, Y. Zeraouli, J. Castaing-Lasvignottes, Experimental determination of the nucleation probability in emulsions, Thermochim. Acta 454 (2007) 57–63.
[28] S. Gibout, M. Strub, J.P. Dumas, Estimation of the nucleation probability in emulsions, Int. J. Heat Mass Transfer 47 (2004) 63–74.
[29] J.P. Dumas, M. Krichi, M. Strub, Y. Zeraouli, Models for the heat transfers during the transformations inside an emulsion-l, crystallizations of the undercooled droplets, Int. J. Heat Mass Transf. 37 (1994) 137–746.
[30] A. Sydow, GMD-FIRST Genetic Algorithms and Genetic Programming Modern Concepts and Practical Applications. Berlin, Germany, 2009.
[31] M. Tembely, R. Attarzadeh. A. Dolatabadi, On the numerical modeling of supercooled micro-droplet impact and freezing on superhydrophobic surfaces, Int. J. Heat Mass Transfer 127 (2018) 193–202.
[32] T. El Rhafiki, T. Kousksou, A. Jamil, S. Jegadheeswaran, S.D. Pohekar, Y. Zeraouli, Crystallization of PCMs inside an emulsion: Supercooling phenomenon, Sol. Energy Mater. Sol. Cells 95 (2011) 2588–2597.
[33] M. Krichi, Etude Des Transferts Thermiques Dans Des Systèmes Dispersés Subissant Des Transformations de Phases (Ph.D. thesis), Univeristy of Pau, 1992.

[34] G. Vallet, D. Trujillo, A stochastic modelling of crystallization in a dispersed medium, Math. Comput. Modelling 36 (2002) 157–172.
[35] I. Karatzas, S.E. Shreve, Brownian Motion and Stochastic Calculus, second ed., Springer-Verlag, New York, 1991.
[36] D. Lamberton, B. Lapeyre, Introduction To Stochastic Calculus Applied To Finance, second ed., in: Chapman and Hall/CRC Financial Mathematics Series, 2011.
[37] P.L. Lions, P.E. Souganidis, Fully nonlinear stochastic partial differential equations, C. R. Math. Acad. Sci. Paris 326 (1998) 1085–1092.
[38] W. Grecksch, C. Tudor, Stochastic Evolution Equations, Akademie Verlag, Berlin, 1995.
[39] F. Black, M. Scholes, The pricing of options and corporate liabilities, J. Political Econ. 81 (3) (1973) 637–654.
[40] E.A. Hussain, Y.M. Alrajhi, Accelerated Genetic Algorithm Solution of Linear Black–Scholes Equation, IJMCAR, 2014.
[41] G.E.P. Box, M.E. Muller, A note on the generation of random normal deviates, Ann. Math. Stat. 29 (2) (1958) 610–611.
[42] J.S.V.R. Krishna Prasad, P.P. Vijay, Finite volume numerical grid technique for solving one and two dimensional heat flow problems, Res. J. Math. Stat. Sci. 2 (8) (2014) 4–9.
[43] Z. Michalewicz, Genetic Algorithms+Data Structures = Evolution Program, Springer, Berlin, Germany, 1994.
[44] J.H. Holland, Genetic algorithms and the optimal allocations of trials, SIAM J. Comput. 2 (2) (1975) 88–105.
[45] A. Rezaie, G. Tsatsaronis, U. Hellwig, Thermal design and optimization of a heat recovery steam generator in a combined-cycle power plant by applying a genetic algorithm, Energy 168 (2019) 346–357.
[46] L. Gosselin, M. Tye-Gingras, F. Mathieu-Potvin, Review of utilization of genetic algorithms in heat transfer problems, Int. J. Heat Mass Transfer 52 (9–10) (2009) 2169–2188.
[47] M. Mitchell, Genetic algorithms: An overview, Complexity 1 (1) (1995) 31–39.
[48] K. Baadache, C. Bougriou, Optimisation of the design of shell and double concentric tubes heat exchanger using the genetic algorithm, Heat Mass Transf. 51 (10) (2015) 1371–1381.
[49] D.E. Goldberg, Genetic Algorithms in Search Optimization and Machine Learning, Addison-Wesley Longman Publishing Co. Inc., 1989.